

Vamshi Krishna Bharadwaj Valluri

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Professional Summary

Third-year Computer Science student specializing in Data Science with strong interest in research-oriented problem solving. Experienced in Python-based data analysis, applied statistical reasoning, and exploratory use of ML and LLM-based systems. Actively developing hands-on machine learning skills with a focus on understanding methodologies, assumptions, and limitations.

Education

B.Tech in Computer Science (Data Science) *Aug 2023 - May 2027*
CR Rao Advanced Institute of Mathematics, Statistics and Computer Science (AIMSCS) *(Expected)*

- GPA: 7.6/10 (up to 4th semester)

S R Junior College – Class 12th

2020 - 2022

- Percentage: 89.8%

Technical Skills

Programming Languages: Python, C++, C, Java

Applied ML Exposure: Using existing ML libraries for regression-based modeling, text analysis, and evaluation in guided and exploratory setups

Data Analysis & Statistics: Data preprocessing, exploratory analysis, basic statistical measures, hypothesis testing (applied)

Tools & Libraries: NumPy, pandas, scikit-learn, Matplotlib, Seaborn, Visual Studio Code, Git, Gemini API, Groq API

Projects

Credit Card Spending Pattern Analysis

- Conducted hypothesis testing on user spending patterns to identify statistically significant differences between credit and debit card usage
- Calculated t-statistics using NumPy with 95% confidence level, revealing distinct behavioral patterns in payment methods
- Emphasized statistical reasoning, hypothesis formulation, and interpretation of results under real-world data assumptions

Critical Study and Experimental Exploration of Text-to-Personality Inference (Big-5 OCEAN) – [GitHub](#)

- Studied and critically analyzed the methodology proposed in “Personality Traits in Large Language Models” by Mustafa Safdari et al. for inferring Big-Five (OCEAN) personality traits from text, with emphasis on the theoretical assumptions underlying linguistic-to-psychological trait mapping.
- Carefully examined the methodology proposed in the paper, including data representation, trait scoring mechanisms, and evaluation assumptions
- Implemented and explored the paper’s workflow to understand how personality signals are extracted from text and how trait consistency is assessed
- Critically analyzed limitations of the approach, including sensitivity to prompt phrasing, ambiguity in linguistic cues, and challenges in validating inferred personality traits
- Identified open research questions related to robustness, interpretability, and ethical considerations in automated personality inference

Blockchain Integration with Relational Databases (Research Analysis)

- Researched blockchain integration approaches for enhancing security and data integrity in relational database systems
- Reviewed academic papers on decentralized database architectures and consensus mechanisms
- Analyzed trade-offs between traditional database performance and blockchain-based immutability

Leadership Experience

Batch Representative – CR Rao AIMSCS

Aug 2023 - Present

- Currently serving as the representative for two batches in college, acting as the primary liaison between students and faculty, and ensuring effective communication and issue resolution.

Certifications

Google AI Essentials – [Course Certificate](#) 

2024

Quantum Computing: Algorithms and Applications – [Course Certificate](#) 

2024

Workshop on Artificial Intelligence with Machine Learning – [Workshop Certificate](#) 

2024